

Davis Liu

☎ +1 (647) 280-0894 • ✉ d53liu@uwaterloo.ca • [in Davis Liu](https://www.linkedin.com/in/DavisLiu) • [G davisliu2006](https://github.com/davisliu2006) • davisliu2006.github.io

Education

University of Waterloo, Bachelor of Software Engineering

Waterloo, ON Sept 2024 – Present

Experience

Software Engineering Intern – Faire

Toronto, ON Jan 2026 – Present

- Implemented and scoped new features for the Faire Retailer Android app such as **enhanced search and filter architectures** and new client/server-driven pages using **Jetpack Compose** and **Kotlin**
- Proposed and implemented **backend** and **SDUI** changes in Kotlin to enhance server-driven app and web functionality
- Wrote automated **unit tests** for **data pipelines**, created test specifications, and orchestrated manual testing parties

Full-Stack Software Engineering Intern – Paige - Kilobryte

Remote, ON May 2025 – Aug 2025

- Developed Paige, a simplified video calling platform for seniors, from a prototype stage to first product launch
- Implemented dedicated device and cross-platform mobile app using **React Native** and **Firestore** database
- Created and maintained 10+ backend endpoints using **NextJS**, connecting external APIs for video calling and notifications

Programming Lead – Unionville High School VEX Robotics Team

Markham, ON Sept 2022 – June 2024

- Led software development** for operator-controlled and **fully autonomous** driving on **3 independent subteams**
- Developed a modular and adaptable **C++ library** for **autonomous feedback control**, on-screen selection menus, and debugging, **reducing iteration time by 75+%** through **6 competition events over 2 years**
- Mentored 10+ junior members** in embedded software development, mechanical design, and CAD

Projects

Programming Language Interpreter  (C++)

Apr 2026

- Built a custom code interpreter in C++ including a syntax parser, VM instruction generator, and memory model
- Supported expression evaluation, control structures, recursive functions, and basic compile-time optimizations

Simple Online Programming Judge  (Docker, Express, MariaDB, TypeScript, EJS/CSS)

Mar 2025 – May 2025

- Built an **online judge** that evaluates code correctness and speed in containerized **Docker-based sandbox environments**
- Implemented web backend using **Express** and databases for user accounts and problems using **MariaDB**
- Deployed web server on a local virtual machine with security measures such as **JWT** and **CAPTCHA**

Chess Engine  (C++)

Jan 2025 – Feb 2025

- Developed an automated **decision-making algorithm** for chess games using **minimax optimization** techniques in C++
- Leveraged **multithreading** to **improve performance by 6.4 times** compared to a single-threaded algorithm
- Obtained a benchmark of **greater than 1000 ELO** on Chess.com with a maximum compute time of **0.16s per move**

DoorSense  (Raspberry Pi, TensorFlow, React Native, Express, TypeScript, Python)

Oct 2024 – Dec 2024

- Collaboratively built a **smart door system** that supports doorstep monitoring, weather displays, and mobile notifications
- Trained custom **TensorFlow convolutional vision models** to identify visitors, package deliveries, and potential intruders
- Integrated system with a **React Native mobile app** using HTTP requests to control smart door functionality

Skills

Languages: C++, TypeScript, JavaScript, Java, Kotlin, C, Python, HTML/CSS, Assembly, SQL

Technologies: Docker, TensorFlow, Express, NextJS, React Native, Electron, MySQL, MongoDB, SDL2, CAD

Awards

Euclid Mathematics Contest - Distinction

Score within the top 14% in 2023

2023, 2024

Canadian Computing Competition - Distinction

Score within the top 6.5% in 2023

2023, 2024